A finite nonreciprocal counterexample to a noise-matched current large-deviation bound

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Abstract

An established thermodynamic bound on the variance of a particle current in a coherent conductor with broken reciprocity suggests a corresponding extension of the reciprocal large-deviation bound. We formulate the proposed extension by matching its curvature to that variance inequality and give a computer-assisted counterexample. The conductor is a stationary Hermitian 387-dot sample attached to three normal tight-binding leads and three fixed Fermi reservoirs at a common positive temperature. Its contact self-energies, finite filter width and full propagating band are retained exactly. At the interior reversal current $j = -J$, a complete interval integration and a fixed-field Fenchel argument establish a rate-function violation greater than $0.019\varepsilon/h$, where $\varepsilon = k_{\mathrm{B}}T_{\mathrm{ref}}$. A conservative analytic bound on the exact dissipation constant removes any reliance on its numerical minimizer. The arithmetic verification includes an exhaustive check of every Taylor remainder used in the fixed certificate, after an intermediate magnitude-bound defect was identified. The result excludes this particular noise-matched single-current shape; it leaves the reciprocal theorem, the nonreciprocal variance bound and the alternative joint-current question intact.

1 The proposed extension and its scope

Mean dissipation constrains fluctuations in coherent fermionic transport, but the constraint depends on the observable and on reciprocity. Brandner and Saito proved a thermodynamic upper bound on the particle-current rate function when the transmission coefficients are symmetric [1]. Their separate variance inequality applies also when this symmetry is broken [2]. In the perspectives section of the final large-deviation paper, they propose inserting the nonreciprocal dissipation factor into the full bound so as to recover the variance inequality. They also raise a distinct possibility involving the joint rate function of all independent currents.

The first suggestion is qualitative. We therefore state explicitly the curvature-matching interpretation tested here. Let $J > 0$ be an oriented single-terminal mean particle current and let $\sigma > 0$ be the actual total reservoir entropy-production rate. Write

$$R_\psi = \coth\!\left(\frac{\sigma}{2\psi k_{\mathrm{B}} J}\right), \qquad c_\psi = JR_\psi, \tag{1}$$

$$B_\psi(j) = (c_\psi + j)\log\frac{c_\psi + j}{c_\psi + J} + (c_\psi - j)\log\frac{c_\psi - j}{c_\psi - J}. \tag{2}$$

The assertion is

$$I(j) \le B_{\psi_0}(j), \qquad -c_{\psi_0} < j < c_{\psi_0}, \tag{3}$$

for every stationary elastic noninteracting normal coherent conductor, every finite number of leads and channels, and all fixed finite reservoir chemical potentials and positive temperatures for which the steady integrals exist. No transmission reciprocity is imposed. Equation (3) is our precise formalization of the first suggestion, rather than a formula quoted verbatim as an established result.

The constant is the exact infimum defined in the supplemental material of Ref. [2]:

$$\psi_0 = \inf_{\substack{0<x,y<1\\ x\neq y}} \frac{\rho(x)-\rho(y)-\rho'(x)(x-y)}{(x-y)\,\operatorname{arsinh}\big((x-y)/(x+y-x^2-y^2)\big)}, \quad \rho(x) = -x\log x-(1-x)\log(1-x). \tag{4}$$

Continuous limiting values are understood. We do not replace this infimum by a rounded decimal. If $C_2 = \chi''(0)$ is the count-variance rate, the published nonreciprocal noise inequality reads

$$\frac{C_2}{J}\sinh\left(\frac{\sigma}{\psi_0 k_{\mathrm{B}} J}\right) \geq 1. \tag{5}$$

Indeed, $B''_\psi(J) = \sinh[\sigma/(\psi k_{\mathrm{B}} J)]/J$, so (1) gives the curvature matching in this ansatz. The reciprocal theorem instead uses $R = \coth[\sigma/(4k_{\mathrm{B}}J)]$; it is not being tested outside its hypotheses.

Our counterexample uses all reservoir heats in σ. This matters because the proved nonreciprocal bound involving a partial dissipation in Ref. [1] does not permit that quantity to be replaced by total dissipation without the additional symmetry premise. Bounds on arbitrary-current precision based on forward-backward event entropy [3] use another input and do not supply (3). A bounded primary-literature and version check on 13 October 2026 identified no later resolution of the precise assertion above. This is not an exhaustive priority claim.

2 Counting conventions and a decisive fixed-field test

At energy E, S_{ab} maps the incident channel of lead b to the outgoing channel of lead a, and $\mathcal{T}_{ab} = |S_{ab}|^2$ in our one-channel sample. The extracted-particle current and the heat injected into the sample are

$$J_a = \frac{1}{h}\int_0^\infty \mathrm{d}E \sum_b \mathcal{T}_{ab}(f_a - f_b), \tag{6}$$

$$Q_a = \frac{1}{h}\int_0^\infty \mathrm{d}E \sum_b \mathcal{T}_{ab}(f_a - f_b)(E-\mu_a), \qquad \sigma = -\sum_a \frac{Q_a}{T_a}, \tag{7}$$

$$f_a(E) = \frac{1}{1+\exp[(E-\mu_a)/(k_{\mathrm{B}}T_a)]}. \tag{8}$$

Here h is Planck's constant. For $F = \mathrm{diag}(f_a)$ and the channel projector V_0, the long-time cumulant-generating rate and its Legendre–Fenchel transform are

$$\chi_0(s) = \frac{1}{h}\int_0^\infty \mathrm{d}E\, \log\det[\mathbb{I} - F + FS^\dagger e^{-sV_0}Se^{sV_0}], \tag{9}$$

$$I_0(j) = \sup_{s\in\mathbb{R}}\{sj - \chi_0(s)\}. \tag{10}$$

Thus s is dimensionless, J, χ, I have units of inverse time and σ has units of entropy per time. There is no extra factor of two from an electrical-noise spectral-density convention.

For one counted channel the energy-resolved count X takes values $-1, 0, 1$. Direct conditioning on the incoming occupation of that channel, or the Slater-determinant scattering probabilities, gives

$$p_+ = f_0\sum_{b\neq 0}\mathcal{T}_{0b}(1-f_b), \qquad p_- = (1-f_0)\sum_{b\neq 0}\mathcal{T}_{0b}f_b, \qquad p_0 = 1-p_+-p_-, \tag{11}$$

$$M(s) = p_0 + p_+e^s + p_-e^{-s}. \tag{12}$$

These probabilities reduce the determinant in (9); they are not an independent stochastic replacement of the scattering model.

The reversal current gives a particularly economical test. For $J, \sigma > 0$ and finite positive ψ, $c_\psi > J$ and direct substitution yields

$$B_\psi(-J) = 2J \log \frac{c_\psi + J}{c_\psi - J} = \frac{2\sigma}{\psi k_{\mathrm{B}}}. \tag{13}$$

Appendix A proves the exact conservative bound $1/2 \le \psi_0 \le 1$. Fenchel's inequality at the fixed field $s = -24$ therefore implies

$$I_0(-J) - B_{\psi_0}(-J) \ge 24J - \chi_0(-24) - \frac{4\sigma}{k_{\mathrm{B}}} \equiv \Delta. \tag{14}$$

A positive enclosure of Δ, together with $J, \sigma > 0$, proves a strict rate-function violation at an interior current. No numerical saddle, rounded dissipation constant or assumption about equality in (10) is required.

3 A finite stationary normal conductor

Fix any $T_{\mathrm{ref}} > 0$ and use the energy unit $\varepsilon = k_{\mathrm{B}} T_{\mathrm{ref}}$. Each of three identical normal one-channel semi-infinite leads has onsite $2t = 20000\varepsilon$ and hopping $-t = -10000\varepsilon$. Its propagating band is $0 < E < 40000\varepsilon$. The retarded surface Green function is

$$g_{\mathrm{lead}}(E) = \frac{E - 2t - \mathrm{i}\sqrt{E(4t - E)}}{2t^2}, \qquad 0 < E < 4t, \tag{15}$$

with the retarded analytic continuation outside the band. It follows from the semi-infinite surface recurrence and equivalently from the outgoing Bloch solution $E = 2t - 2t\cos k$.

The sample consists of three identical $n = 128$ site filter chains and three core sites, hence $3n + 3 = 387$ sites. Set

$$\begin{aligned} &E_0 = 5000\varepsilon, && w = 0.02\varepsilon, && \Gamma_c = 200000\varepsilon, \\ &\Gamma_f = \frac{w}{4\sin[\pi/(2n)]}, && h_\ell = \frac{w}{4\sqrt{\sin[(2\ell+1)\pi/(2n)]\sin[(2\ell-1)\pi/(2n)]}}, \\ &\lambda = \sqrt{\Gamma_f \Gamma_c}, && 1 \le \ell \le n-1. \end{aligned} \tag{16}$$

The filter profile uses the $w/4$ convention of the quantum-dot construction in Ref. [2]. These equations, rather than any approximate transmission-window description, define the sample.

Label the filter sites (a, ℓ) and the core sites c_a, with $a = 0, 1, 2$. The filter hoppings are the real h_ℓ. The real hopping λ joins (a, n) to c_a. All sample onsite energies are E_0, except that of the outer site $(a, 1)$, which is $E_0 - \mathrm{Re}\,\Sigma(E_0)$. The real lead contact is defined by

$$V^2 = \frac{2t^2\Gamma_f}{\sqrt{E_0(4t - E_0)}}, \qquad \Sigma(E) = V^2 g_{\mathrm{lead}}(E). \tag{17}$$

On the three core sites the remaining hopping matrix is

$$H_{\mathrm{core}} - E_0 \mathbb{I} = \mathrm{i}\Gamma_c (P - P^{\mathsf{T}}), \qquad P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \tag{18}$$

It is Hermitian and carries a nonzero magnetic flux. All hoppings, contacts and onsite energies are fixed and time independent. There are no interactions, anomalous pairing terms, drives, probes or internal baths.

For $0 < E < 4t$, let $\Gamma(E) = -\,\mathrm{Im}\,\Sigma(E) > 0$ and

$$G(E) = [E\mathbb{I} - H_{\mathrm{sample}} - \Sigma(E) P_{\mathrm{out}}]^{-1}, \qquad S_{ab}(E) = \delta_{ab} - 2\mathrm{i}\Gamma(E) G_{(a,1),(b,1)}(E), \tag{19}$$

where P_{out} projects onto the three outer filter sites. The resolvent identity $G-G^\dagger = -2\mathrm{i}\Gamma G P_{\mathrm{out}} G^\dagger$ proves unitarity of this flux-normalized scattering matrix. Outside the propagating band the leads have no open channels, so there is no reservoir-current contribution. Discrete bound sample modes carry no extensive steady reservoir current in this stationary noninteracting model. The lead endpoints can be handled by their limiting scattering values and have zero integration measure.

The three reservoirs remain at the common temperature T_{ref} and at the fixed chemical potentials

$$(\mu_0, \mu_1, \mu_2) = (5001, 4997, 4976)\varepsilon. \tag{20}$$

No energy-dependent occupation is chosen independently of these Fermi functions. The selected extracted-particle current is already positive; no orientation reversal is applied.

4 Total entropy, circulation and the integrated certificate

The sample is circulant in its lead indices. Let $u = \mathcal{T}_{01}$, $v = \mathcal{T}_{02}$, $d_{01} = f_0 - f_1 > 0$ and $d_{12} = f_1 - f_2 > 0$. The two circulation sectors contribute respectively the current-density vectors

$$u(d_{01}, d_{12}, -d_{01} - d_{12}), \qquad v(d_{01} + d_{12}, -d_{01}, -d_{12}). \tag{21}$$

Their sum conserves particle number. At the common temperature, conservation of total energy and particle number converts the heat definition (7) into

$$\frac{\sigma}{k_{\mathrm{B}}} = \sum_a \frac{\mu_a - E_0}{\varepsilon} J_a. \tag{22}$$

In particular, define the dimensionless energy densities $\mathcal{J}$ and $\mathcal{E}$ by

$$\mathcal{J} = u d_{01} + v(d_{01} + d_{12}), \tag{23}$$

$$\mathcal{E} = u(25d_{01} + 21d_{12}) + v(4d_{01} + 25d_{12}). \tag{24}$$

Their energy integrals divided by h give J and σ/k_{B}. Both are nonnegative, and the central circulating transmission makes them strictly positive on a positive-width set. All three heats are included in (24).

For numerical evaluation use $q = (E/\varepsilon - 5000)/0.01$, with $\mathrm{d}E = 0.01\varepsilon \mathrm{d}q$. Put $z = e^{0.01q}$, $e_0 = e^{-1}$, $e_1 = e^3$, $e_2 = e^{24}$ and $D_a = 1 + z e_a$. Then

$$d_{01} = \frac{z(e_1 - e_0)}{D_0 D_1}, \qquad d_{12} = \frac{z(e_2 - e_1)}{D_1 D_2}, \qquad A_b = \frac{z[e_b(e^{-24} - 1) + e_0(e^{24} - 1)]}{D_0 D_b}. \tag{25}$$

Equations (11)–(12) give $M(-24) = 1 + uA_1 + vA_2$. The residual in (14) is integrated directly with density

$$\mathcal{D} = u(-76d_{01} - 84d_{12}) + v(8d_{01} - 76d_{12}) - \log(1 + uA_1 + vA_2). \tag{26}$$

This retains correlations between current and entropy. Subtracting loose separate observable enclosures would yield a weaker result.

The full finite-Hamiltonian range construction is described in Appendices B and C. It covers $q \in [-8192, 16384]$, corresponding to $E/\varepsilon - 5000 \in [-81.92, 163.84]$, using exactly 26,275 consecutive dyadic panels. Range bounds apply to every energy in a panel, including resonances. All 727 panels with zero-inclusive interval denominator bounds are retained using the physical transmission bounds. Analytic Fermi tails enclose the rest of the actual propagating band. The tail contribution to the absolute residual uncertainty is below $1.12 \times 10^{-14}\varepsilon/h$.

Table 1 gives conservative outward decimal enclosures obtained by exact-rational summation of the exported panel bounds and directed tail estimates. They are intervals, not asserted

Table 1: Outward enclosures for the fixed physical sample. All entries are in units of ε/h. The residual is enclosed directly, rather than by subtracting the other rows.

Quantity	Lower endpoint	Upper endpoint
Δ	0.019447	0.024248
J	0.013576	0.014625
σ/k_{B}	0.073446	0.079193
$\chi_0(-24)$	0.010467	0.012105

significant digits of a point estimate. More detailed exact endpoints are supplied with the reproducibility data. In particular,

$$\Delta \in [0.019447591396978867,\, 0.024247460886202978]\,\frac{\varepsilon}{h}. \tag{27}$$

The enclosure width is below $0.005\varepsilon/h$.

Proposition 1. *For the stationary sample and reservoirs in Section 3, the curvature-matched assertion (3) is false. In the stated computer-assisted arithmetic trust model,*

$$I_0(-J) - B_{\psi_0}(-J) > 0.019\,\frac{\varepsilon}{h}. \tag{28}$$

Proof. The complete certificate proves $J, \sigma > 0$ and $\Delta > 0.019\varepsilon/h$. Appendix A gives a positive finite exact ψ_0, so $R_{\psi_0} > 1$ and $-J$ is strictly inside the asserted domain. Equation (14) proves the strict violation. One admissible finite sample suffices to refute the universal quantifiers. □

5 Verification, physical meaning and limitations

The integral proof uses complete range coverage, rather than quadrature convergence. An independently implemented check reconstructs the retarded boundary and branches by outgoing Bloch modes and scalar continuants; another uses forward branch elimination and a direct core inverse. Explicit Slater minors agree with the original counting determinant. The latter implementation checks 58 high-precision scattering/counting points, 41 comparisons with full-panel bounds and five dense 387-site resolvent solves. The maximum dense-solve scattering disagreement is below 5.6×10^{-15}. Gauss orders 2, 4 and 8 on the frozen partition give a diagnostic residual near $0.02382794\varepsilon/h$, with order-4-to-8 difference about $8.1\times10^{-9}\varepsilon/h$. These agreements are diagnostics, not the source of the strict integral error bound. A positive-temperature reciprocal boxcar baseline also reproduces the published narrow-window current and CGF limits. Parameters, primary inputs and arithmetic libraries are shared between these implementations; code and scattering values are not reused for the independent physical reconstruction.

The range verification has an explicit qualification. An intermediate scalar absolute-value helper is nearest rounded and fails to be an outward magnitude bound for 42 actual coefficients. The fixed calculation is validated by an independent directed 500-digit reconstruction and an exhaustive comparison of every implemented remainder upper endpoint: 52,550 checks, two polynomials on all 26,275 final panels, with zero violations. Later directed arithmetic provides sufficient margin for these actual remainders. This establishes the fixed certificate, not generic outward correctness of the helper for other grids or parameters. The failure records and complete closure checker are retained with the data.

The mechanism can be read directly from (21). Around the central circulating transmission, particles flow from the high-chemical- potential reservoir into the intermediate reservoir and then toward the low-chemical-potential reservoir. The intermediate reservoir reduces the total mean entropy cost per selected extracted particle. In contrast, a negative selected count is sensitive

to the occupation of the much lower- chemical-potential reservoir. The large negative counting field accentuates that rare event. Finite filtering and magnetic flux implement this separation in a physical stationary sample. Leakage at the filter edges is not negligible by assumption; it is included in the certificate.

Accordingly, matching the curvature of a variance inequality does not determine the proposed full rare-current shape from actual mean entropy. The result says nothing against the established reciprocal large-deviation theorem or the nonreciprocal variance inequality. It does not resolve the joint-current alternative, exclude other total-entropy-dependent shapes, or assert that the sample has been fabricated. The filter order, width, core coupling and lead band are fixed. There is no infinite-size, zero-temperature, wide-band or perturbative limit in the proof.

The computation trusts Python exact integer/rational arithmetic, mpmath 1.3.0 directed interval elementary functions and nearest-rounded arbitrary-precision basic arithmetic with the explicit forward-error envelope. It is a computer-assisted model counterexample, without proof-assistant formalization. The numerical margin is conditional on that documented computational trust base.

Computational research method. OpenAI Codex assisted literature discovery, model selection, the analytic reduction, code generation, computational checking and interpretation. The reported intervals are outputs of the deterministic scattering and range calculations described above. Separate implementations and the complete fixed-certificate remainder comparison supply the disclosed validation; AI text or confidence is not a numerical error estimate.

Data availability. The accompanying local reproducibility package contains the defining parameters, exact-model derivations, complete exported panel data, calculation scripts, arithmetic-defect records and the exhaustive remainder-check code and outputs. It includes instructions for scratch-copy recomputation and exact resummation. No public repository or deposited-data identifier is claimed for this draft.

Acknowledgements and AI disclosure

OpenAI Codex was used for research ideation, literature discovery, analytic reasoning, computational implementation and checks, and manuscript drafting. The disclosed validation consists of deterministic calculations, complete range/error checks and separate AI-assisted implementations. Human scientific review, critical revision of the AI-assisted text and references, and confirmation of authorship remain pending for this local draft.

A An analytic lower bound for the exact constant

The numerator in (4) is the binary relative entropy $D(y\|x)$. Jensen's inequality applied to the square-root likelihood ratio gives

$$D(y\|x) \geq -2\log C, \qquad C = \sqrt{xy} + \sqrt{(1-x)(1-y)}. \tag{29}$$

Put $x = \sin^2\alpha$, $y = \sin^2\beta$ with $0 < \alpha, \beta < \pi/2$. First take $\alpha > \beta$, and define $r = \sin(\alpha-\beta)$, $A = \sin(\alpha+\beta)$, so $0 < r \leq A \leq 1$. Then

$$C = \sqrt{1-r^2}, \qquad x - y = Ar, \qquad x + y - x^2 - y^2 = \tfrac{1}{2}[A^2(1-2r^2) + r^2]. \tag{30}$$

For fixed r, let

$$F_r(A) = Ar \operatorname{arsinh} \frac{2Ar}{A^2(1-2r^2) + r^2}. \tag{31}$$

With $v = 2Ar/[A^2(1-2r^2)+r^2] > 0$ and $b = 1-2r^2$, differentiation gives

$$\frac{F_r'(A)}{r} = \operatorname{arsinh} v + \frac{v}{\sqrt{1+v^2}}\frac{r^2 - bA^2}{r^2 + bA^2} \geq \operatorname{arsinh} v - \frac{v}{\sqrt{1+v^2}} \geq 0. \tag{32}$$

The first inequality follows because the displayed rational factor is at least -1; the second follows by differentiation from $v = 0$. Thus the denominator of (4) is at most

$$F_r(1) = r \operatorname{arsinh} \frac{2r}{1-r^2} = 2r \operatorname{artanh} r. \tag{33}$$

Also $g(r) = -\log(1-r^2) - r\operatorname{artanh} r$ satisfies $g(0) = 0$ and

$$g'(r) = \frac{r}{1-r^2} - \operatorname{artanh} r \geq 0, \tag{34}$$

since the derivative of the right-hand side is $2r^2/(1-r^2)^2 \geq 0$ and its value at zero is zero. Combining the bounds,

$$D(y\|x) \geq -\log(1-r^2) \geq r\operatorname{artanh} r \geq \tfrac{1}{2}(x-y)\operatorname{arsinh}\frac{x-y}{x+y-x^2-y^2}. \tag{35}$$

The square-root likelihood bound holds in both relative-entropy directions; C and the denominator are symmetric. The argument therefore covers $x < y$ as well as all continuous limits, proving $\psi_0 \geq 1/2$. At $y = x + \delta$ the numerator and denominator both have leading term $\delta^2/[2x(1-x)]$, so the diagonal limit is one and $\psi_0 \leq 1$.

B Finite continuants and transmission ranges

This appendix uses 0.01ε as the energy scale, so the filter diagonal variable is q. Define

$$\begin{aligned} \gamma_f &= \frac{1}{2\sin(\pi/256)}, \quad g_c = 20000000, \quad \lambda_{\mathrm{q}}^2 = \gamma_f g_c, \\ b_\ell &= \frac{1}{4\sin[(2\ell+1)\pi/256]\sin[(2\ell-1)\pi/256]}, \end{aligned} \tag{36}$$

where λ_{q} denotes the dimensionless core-filter hopping. The closed-chain continuant has recurrence $P_{j+1} = qP_j - b_j P_{j-1}$, $P_0 = 1$, $P_1 = q$. Let F be the 128-site determinant, B the 127-site determinant with the first site removed and G the 126-site determinant with both endpoints removed. The profile is reflection symmetric, so deleting the other endpoint gives the same B.

For clarity define

$$a_c = \frac{\Gamma_f/\varepsilon}{\sqrt{5000\cdot 35000}}, \qquad \gamma(q) = \gamma_f\sqrt{(1+0.01q/5000)(1-0.01q/35000)}. \tag{37}$$

The exact contact-modified determinants are

$$D = F + (\mathrm{i}\gamma - a_c q)B, \qquad C = qD - \lambda_{\mathrm{q}}^2[B + (\mathrm{i}\gamma - a_c q)G]. \tag{38}$$

The core Schur complement is $(C/D)\mathbb{I} - \mathrm{i}g_c(P - P^{\mathsf{T}})$. Its three sector determinants are C, $C - \sqrt{3}g_c D$ and $C + \sqrt{3}g_c D$. Put $H = \prod_{\ell=1}^{127} b_\ell$ and

$$\mathcal{A} = \frac{4\gamma^2(\lambda_{\mathrm{q}}^2)^2 H^2 g_c^2}{|C|^2|C - \sqrt{3}g_c D|^2|C + \sqrt{3}g_c D|^2}. \tag{39}$$

The endpoint cofactors and core adjugate give

$$u = \mathcal{A}|C - \mathrm{i}g_c D|^2, \qquad v = \mathcal{A}|C + \mathrm{i}g_c D|^2. \tag{40}$$

These identities refer to the finite physical Hamiltonian. At $q = 0$ they give $u = 0$, $v = 1$, consistent with the current orientation above.

For the small reverse numerator the cancellation is performed before interval evaluation. With $B_c = qB - \lambda_q^2 G$ and $K = F + \gamma_f^2 G$, the exact components are

$$\mathrm{Re}(C - \mathrm{i}g_c D) = qF + g_c(\gamma - \gamma_f)B - a_c q B_c, \tag{41}$$

$$\mathrm{Im}(C - \mathrm{i}g_c D) = q(\gamma + g_c a_c)B - g_c(\gamma/\gamma_f)K + g_c(\gamma/\gamma_f - 1)F. \tag{42}$$

All coefficients of K, including small nonleading ones, are retained. No inferred zero, approximate filter identity or wide-band replacement is used. If an interval denominator contains zero, the panel uses $0 \le u, v \le 1$ from unitarity; it is not removed. The separate physical bound $e^{-24} \le M(-24) \le e^{24}$ ensures a defined logarithm.

C Whole-panel arithmetic and the fixed-certificate closure

The trigonometric hoppings and continuant coefficients are generated with directed interval arithmetic at 250 decimal digits. Write $F(q) = F_e(q^2)$ and $B(q) = qB_e(q^2)$. On a dyadic q panel let $x = q^2$, $x_c = m^2$ for its midpoint m, and $r_x \ge \sup|x - x_c|$, $X \ge \sup|x|$. A degree-19 Taylor range for either even polynomial $p(x) = \sum_k p_k x^k$ includes the remainder

$$\mathcal{R}_p \le r_x^{20} \sum_{k \ge 20} |p_k| \binom{k}{20} X^{k-20}. \tag{43}$$

This bounds all omitted degrees throughout the panel.

The first 20 Taylor coefficients are evaluated by generalized Horner jets using 336-bit nearest-rounded point arithmetic. The dyadic midpoint square is exactly representable. Let $u_p = 2^{1-336}$, $N = 5544$ and $\gamma_N = Nu_p/(1 - Nu_p)$. The operation count exceeds all basic operations in each signed or positive jet recurrence. If $\widehat{A}$ is the computed positive recurrence for absolute coefficient sums and r_c is the directed relative coefficient uncertainty, the jet envelope is

$$\widehat{\text{jet}} \ \pm \ \frac{(\gamma_N + r_c)\widehat{A}}{1 - \gamma_N}. \tag{44}$$

It includes cancellation and rounding of the positive recurrence. Subsequent Taylor evaluation, norms, divisions, square roots, exponentials, logarithms, panel widths and sums use directed 85-digit interval operations.

For reproducibility it is essential to describe a defect in the original remainder implementation. The scalar absolute-value conversion used to initialize coefficient magnitudes is nearest rounded. Independent exact endpoint comparisons show 44 failures to enclose the original coefficient intervals, including 42 strict underestimates of the exact coefficient magnitudes. This helper cannot be treated as an outward upper-bound operation in general.

The closure is specific and complete. A separately transcribed checker reconstructs coefficient intervals at 500 digits. It extracts absolute endpoint bounds by changing their encoded sign, avoiding nearest-rounded scalar absolute values. At every actual final panel, it evaluates a 500-digit directed reference bound for (43) and compares it to the implemented 85-digit remainder upper endpoint. Each implemented upper endpoint is at least the reference upper endpoint, for both polynomials on every one of the 26,275 panels. All 52,550 comparisons complete with zero violations. Hence every remainder used in this fixed certificate is adequate despite the defective intermediate initialization. The supplemental code and records preserve both the failure and this exhaustive closure; they do not certify a reusable helper for new inputs.

Every panel supplies a range for the full density. Its positive width times that range encloses the integral, including structures narrower than the panel. The refinement driver can retain

valid parent intervals or replace them by child ranges. The final consecutive dyadic endpoints and width subtractions are checked as exact rationals, with total q width 24576. Each binary64 exported lower/upper endpoint is moved one representable number outward. Exact-rational resummation covers every exported panel, including all 727 denominator fallbacks, and adds the tails below once.

D Finite-band tails

Let $\delta = E/\varepsilon - 5000$. Below -81.92, every reservoir hole probability is bounded by $e^{\delta+24}$; above 163.84, every particle probability is bounded by $e^{1-\delta}$. A nonzero extracted count requires at least one hole in the low-energy tail or at least one particle in the high-energy tail. Conservative dimensionless integrated event bounds are

$$H_t = 3e^{-81.92+24}, \qquad P_t = 3e^{1-163.84}. \tag{45}$$

Throughout these tails the e^{24}-amplified event probability is below $1/2$. From $|M-1| \le e^{24}\Pr(X \neq 0)$ and $|\log M| \le 2|M-1|$, the following bounds hold in rate units ε/h:

$$J_{\mathrm{tail}} \le 2(H_t + P_t), \tag{46}$$

$$(\sigma/k_{\mathrm{B}})_{\mathrm{tail}} \le 28 J_{\mathrm{tail}}, \tag{47}$$

$$|\chi(-24)|_{\mathrm{tail}} \le 2e^{24}(H_t + P_t), \tag{48}$$

$$|\Delta|_{\mathrm{tail}} \le 24 J_{\mathrm{tail}} + 4(\sigma/k_{\mathrm{B}})_{\mathrm{tail}} + |\chi(-24)|_{\mathrm{tail}}. \tag{49}$$

Directed exponentials bound the last expression by $1.1140 \times 10^{-14}\varepsilon/h$. Extending the estimates beyond the actual lead band only enlarges the bound. Together with the complete central partition this covers all physical energies; no infinite-energy truncation is left uncontrolled.

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